Set Relations

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What is Relation

Let A and B be sets. A binary relation R is a subset of A x B

- Example:

A = {1,2} and B = {1,2,3}

A x B = (1,1) (1,2) (1,3) (2,1) (2,2) (2,3)

R = { (a,b) € A x B | a < b}

R ?

R = (1,2) (1,3) (2,3)

Types/Properties of Relation

1. Reflexive Relation
2. Symmetric Relation
3. Transitive Relation
4. Anti Symmetric Relation
5. Irreflexive Relation
6. Equivalence Relation
7. Congruence Relation
8. Partial Relation
9. Inverse Relation
10. Complementary Relation
11. Composite Relation

Reflexive and Irreflexive Relation

Reflexive Relation

every element is related to itself

if set A = {a,b} then R = {(a,a), (b,b)} is reflexive relation.

Irreflexive Relation

no element is related to itself

if set A = {a,b} then R = {(a,b), (b,a)} is irreflexive relation.

Reflexive/Irreflexive Relation Example

Let A = {1,2,3,4}, we define different relations A x A
A x A = {(1,1)(1,2) (1,3) (1,4) (2,1) (2,2) (2,3) (2,4) (3,1) (3,2)(3,3) (3,4) (4,1) (4,2)(4,4)(4,4)}

R1 = {(1,1) (2,2) (3,3) (4,4)}

R2 = {(1,1) (1,4) (2,2) (3,3) (4,3)}

R3 = {(1,1) (1,2) (2,1) (2,2) (3,3) (4,4)}

R4 = {(1,3) (2,2) (2,4) (3,1) (4,4)}

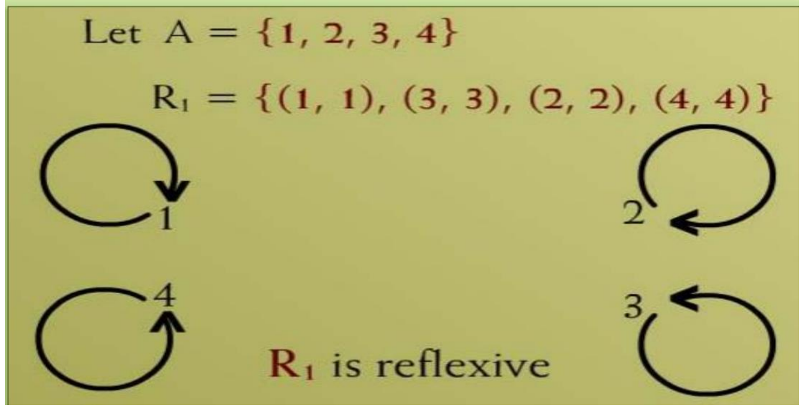
R1 = {(1,3) (1,4) (2,3) (2,4) (3,1) (3,4)}

R2 = {(1,1) (1,2) (2,1) (2,2) (3,3) (4,3)}

R3 = {(1,2) (2,3) (3,3) (3,4)}

Reflexive/Irreflexive Relation Graph Prestation

- Reflexive Relation



- Irreflexive Relation

Reflexive
• The graph of reflexive relation has loop on every element of set A

Irreflexive
• The graph of irreflexive relation has no loop on any • element of set A

Reflexive/Irreflexive Relation Matrix Prestation

- Reflexive Relation

Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,2), (3,3)\}$

R is reflexive

$$M = \begin{array}{c c} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{array}$$

- Irreflexive Relation

$A = \{1,2,3\}$

$R = \{(1,3), (2,1), (2,3), (3,2)\}$

Matrix Representation

$$M = \begin{array}{c c} \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$

R is irreflexive

Solution

R1 = {(1,1) (2,2) (3,3) (4,4)}

R1 is reflexive

R2 = {(1,1) (1,4) (2,2) (3,3) (4,3)}

- R2 is not reflexive, because (4,4) is missing.

R3 = {(1,1) (1,2) (2,1) (2,2) (3,3) (4,4)}

- R3 is reflexive

R4 = {(1,3) (2,2) (2,4) (3,1) (4,4)}

- R4 is not reflexive, because (3,3) is missing.

R1 = {(1,3) (1,4) (2,3) (2,4) (3,1) (3,4)}

R1 is Irreflexive

R2 = {(1,1) (1,2) (2,1) (2,2) (3,3) (4,3)}

- R2 is not **Irreflexive**, because (1,1) (2,2) is.

R3 = {(1,2) (2,3) (3,3) (3,4)}

- R4 is not **Irreflexive**, because (3,3) is missing.

Symmetric and Anti-Symmetric Relation

Symmetric Relation

Relation R on set A is symmetric if (b, a)∈R and (a,b)∈R

An example of symmetric relation will be R = {(1, 2), (2, 1)} for a set A = {1, 2}.

Anti-Symmetric Relation

The relation R is said to be antisymmetric on a set A, if aRb and bRa hold when

a = b

An Example R={(1,2)(2,2)(2,3)(2,4)

R is anti Symmetric

Symmetric and Anti-Symmetric Relation Example

Let A = {1,2,3,4}, we define different relations A x A

A x A = {(1,1)(1,2) (1,3) (1,4) (2,1) (2,2) (2,3) (2,4) (3,1) (3,2)(3,3) (3,4) (4,1) (4,2)(4,4)(4,4)}

R1 = {(1,1)(1,3) (2,4) (3,1) (4,2)}

R2 = {(1,1) (2,2) (3,3) (4,4)}

R3 = {(1,1) (1,2) (2,1) (2,2) (3,3) (4,4)}

R4 = {(1,3) (2,2) (2,4) (3,1) (4,4)}

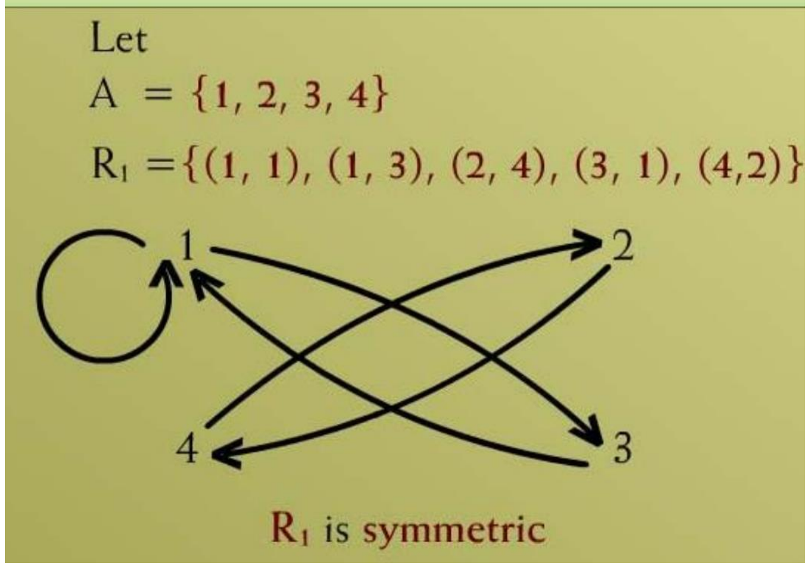
R1 = {(1,1) (2,2) (3,3)}

R2 = {(1,2) (2,2) (2,3) (3,4) (4,1)}

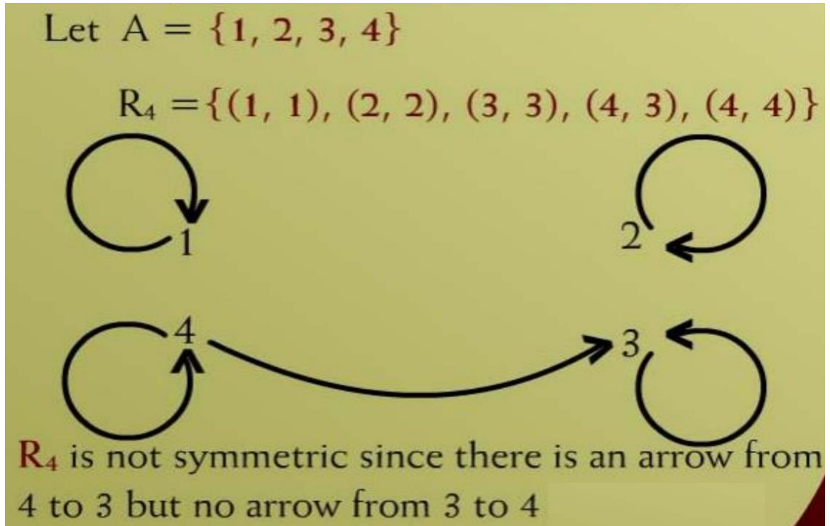
R3 = {(1,3) (2,3) (2,4) (3,1) (4,2)}

Solution

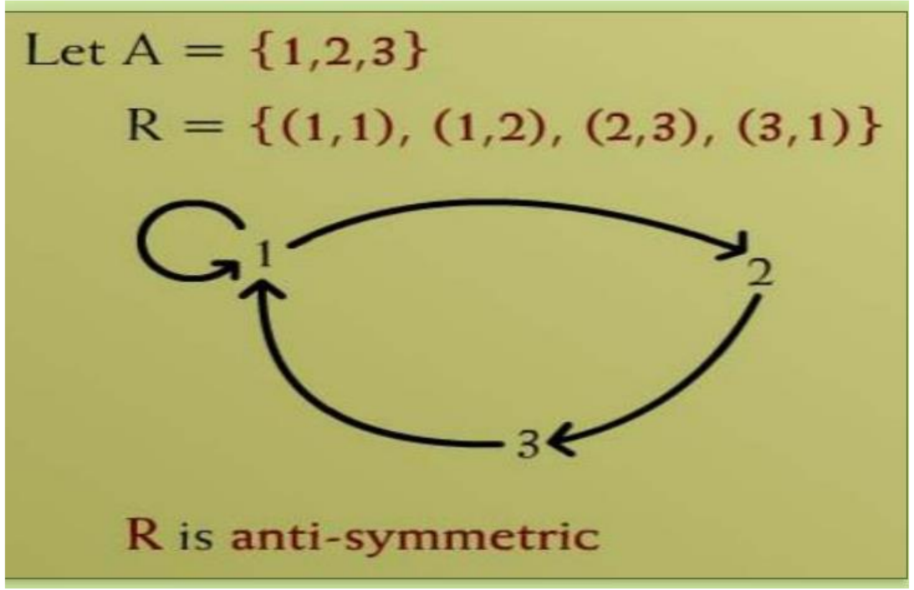
R1 = {(1,1)(1,3) (2,4) (3,1) (4,2)}



R4 = {(1,3) (2,2) (2,4) (3,1) (4,4)}



R1 = {(1,3) (1,4) (2,3) (2,4) (3,1) (3,4)}



Symmetric and Anti-Symmetric Relation Example

Let A = {1,2,3,4}, we define different relations A x A
A x A = {(1,1)(1,2) (1,3) (1,4) (2,1) (2,2) (2,3) (2,4) (3,1) (3,2)(3,3) (3,4) (4,1) (4,2)(4,4)(4,4)}

R1 = {(1,1)(1,3) (2,4) (3,1) (4,2)}

R1 is symmetric

R2 = {(1,1) (2,2) (3,3) (4,4)}

- **R2 is symmetric**

R3 = {(1,1) (1,2) (2,1) (2,2) (3,3) (4,4)}

- **R3 is not symmetric**

R4 = {(1,3) (2,2) (2,4) (3,1) (4,4)}

- **R4 is not symmetric**

R1 = {(1,1) (2,2) (3,3)}

Both Anti-symmetric and symmetric

R2 = {(1,2) (2,2) (2,3) (3,4) (4,1)}

Anti-symmetric but not symmetric

R3 = {(1,3) (2,3) (2,4) (3,1) (4,2)}

Not Anti-symmetric but symmetric

Matrix Representation

Let $A = \{1,2,3\}$

$R = \{(1,3), (2,2), (3,1), (3,3)\}$

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{array}$$

M is symmetric i.e $M = M^t$

Let $A = \{1, 2, 3\}$

$R = \{ (1, 1), (1, 2), (2, 3), (3, 1) \}$

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$$

Transitive Relation

- A relation is said to be a transitive relation, if "a is related to b" and "b is related to c" implies that "a is related to c.

Let R be a relation on a set A.
R is transitive if and only if
for all a, b, c $\in$ A, if (a, b) $\in$ R
and (b, c) $\in$ R then (a, c) $\in$ R.

That is, if aRb and bRc then aRc.

if any one element is related to a second and that second element is related to a third, then the first is related to the third.

Transitive Relation Example

Let A = {1,2,3,4}, we define different relations A x A

A x A = {(1,1)(1,2) (1,3) (1,4) (2,1) (2,2) (2,3) (2,4) (3,1) (3,2)(3,3) (3,4) (4,1) (4,2)(4,4)(4,4)}

R1 = {(1,1)(1,2) (1,3) (2,3)}

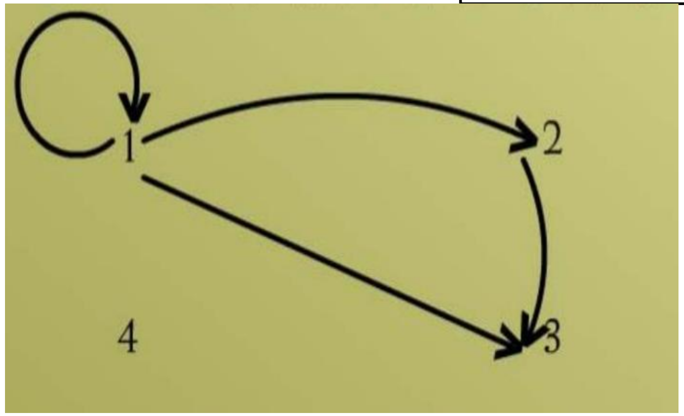
R1 IS TRANSITIVE

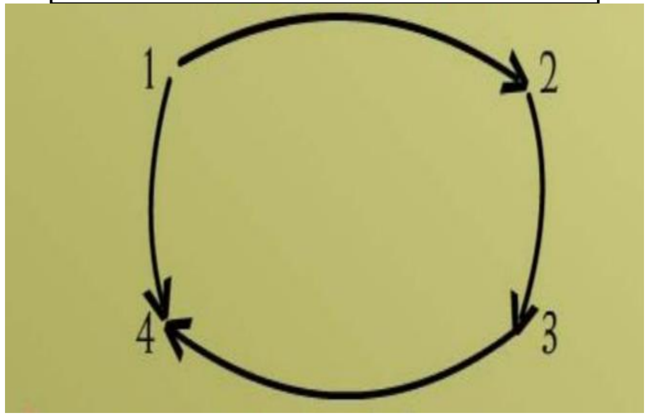
R2 = {(1,2) (1,4) (2,3) (3,4)}

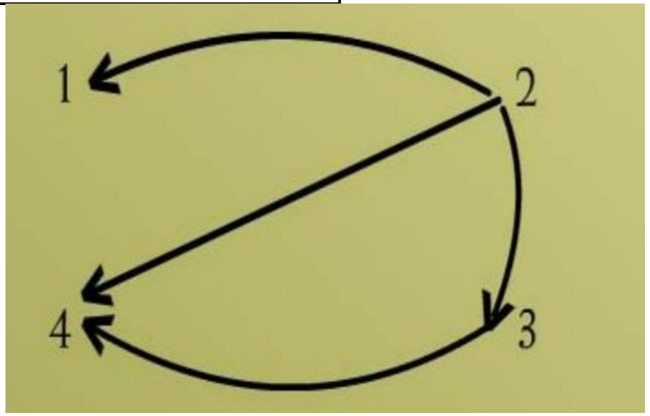
- **R2 IS NOT TRANSITIVE**

R3 = {(2,1) (2,4) (2,3) (3,4)}

- **R3 IS TRANSITIVE**







Examples

Example 1:

Let A = {0, 1, 2}

R = {(0,2), (1,1), (2,0)}

1. Is R reflexive? Symmetric? Transitive?
2. Which ordered pairs are needed in R to make it a reflexive and transitive relation.

Solution

1.R is not reflexive.
R to be reflexive it must contains (0, 0) and (2, 2).

2.R is Symmetric

3. R is not transitive. For R to be transitive it must contain (0, 0) and (2,2)

Example 2:

Define a relation L on the set of real numbers R be defined as follows:

for all x, y $\in$ R, $\quad x\,L\,y \Leftrightarrow x < y$.

a. Is L reflexive?
b. Is L symmetric?
c. Is L transitive?

Solution

1.L is not reflexive.
R to be reflexive (1,1) $1 \nless 1$

2.L is not Symmetric because R must contain (a,b) (b,a) so1 < 2 but $2 \nless 1$.

3. L is transitive. For R to be transitive (1,b) (b,c) and (a,c) if a < b , b < c then a < c.

Example 3:

Define a relation R on the set of positive integers Z^+ as follows:

for all a, b $\in Z^+$, a R b iff $a \times b$ is odd.

Determine whether the relation is.

a. Reflexive

b. Symmetric

c. Transitive

Solution

1.R is not reflexive.
R 1 X 1 = 1 BUT 2 X 2 =4 WHICH IS NOT ODD.

2.R is Symmetric because R must contain (a,b) (b,a) so 1 x 3 =3 then 3 x 1 = 3.

3. R is transitive. If a x b = odd , b x c = odd then a x c iis also odd

EQUIVALENCE RELATION

A be a non-empty set and R a binary relation on A. R is an equivalence relation if and only if, R is reflexive, symmetric, and transitive.

- **Example**
- Let A = {1, 2, 3, 4}
- R = { (1, 1) , (2, 2) , (2, 4) , (3, 3) , (4, 2) , (4, 4) }
- Is R is reflexive, symmetric and transitive.

Congruence Relation

$$m \equiv n \pmod{d}$$

M and N can be integer and d is a positive integer

Is $22 \equiv 1 \pmod{3}$	$22 - 1 = 21$ 21 is divisible by 3	Yes, it is a congruence relation
Is $7 \equiv 7 \pmod{4}$	$7 - 7 = 0$ 0 is divisible by 4.	Yes, it is a congruence relation
Is $14 \equiv 4 \pmod{4}$	$14 - 4 = 10$ 10 is not divisible by 4.	No, it is not a congruence relation

Example:

Define a relation R on the set of all integers Z as follows:
for all integers m and n,

$$m \, R \, n \Leftrightarrow m \equiv n \pmod{3}$$

Prove that R is an equivalence relation.

Solution:

Relation R is equivalence.

Example Proof:

Reflexives	**Symmetric**	**Transitive**
Yes	**Yes**	**Yes**
Reason Reflexive relations are (1,1) ,(2,2) and so on Therefore, m=1 and n=1 $m \equiv n \pmod 3$ $1 \equiv 1 \pmod 3$ $1-1 = 0$ And 0 is dividable by 3.	**Reason** If m R n then n R m, $m R n \rightarrow m \equiv n \pmod 3$ $3 \mid m - n$ $m - n = 3k$ $n - m = -3k$ -3 is also belong to Z So given relation is also symmetric.	**Reason** mRn , nRp $m R n \rightarrow m \equiv n \pmod 3$ $3 \mid m - n$ $m - n = 3k$ (1) $n R p \rightarrow n \equiv p \pmod 3$ $3 \mid n - p$ $n - p = 3c$ (2) By adding eq 1 and 2 $m - n + n - p = 3k + 3s$ $M - p = 3 (k + c)$ (k + c) € Z $m - p = 3q$ $m \equiv p \pmod 3$

Partial Relation

- R be a binary relation defined on a set A. R is a
- partial order relation, if and only if, R is
  - reflexive,
  - anti-symmetric
  - Transitive
- **Example**

R3 = {(1,1)(1,2)(1,3)(1,4)(2,2)(2,3)(2,4)(3,3)(3,4)(4,4)}

Is R3 is partial relation.

Solution R = {(1,1)(1,2)(1,3)(1,4)(2,2)(2,3)(2,4)(3,3)(3,4)(4,4)}

(1,1) (2,2) (3,3)(4,4) belongs to R	R is reflexive
In given there is no relation which aRb and bRa	R is Anti Symmetric
R = (1,2) (2,3) (1,3)	R is transitive

<u>So R is Partial Order Relation</u>

Inverse Relation

An inverse relation is the inverse of a given relation obtained by ***Interchanging or swapping*** the elements of each ordered pair.

<u>if (x, y) is a point in a relation R, then (y, x) is an element in the inverse relation R^{-1}.</u>

Example

- Let A = {2,3,4} and B ={ 2,6,8} find A x B.

***A* x *B* = {(2.2)(2,6)(2,8) (3.2)(3,6)(3,8) (4.2)(4,6)(4,8)}**

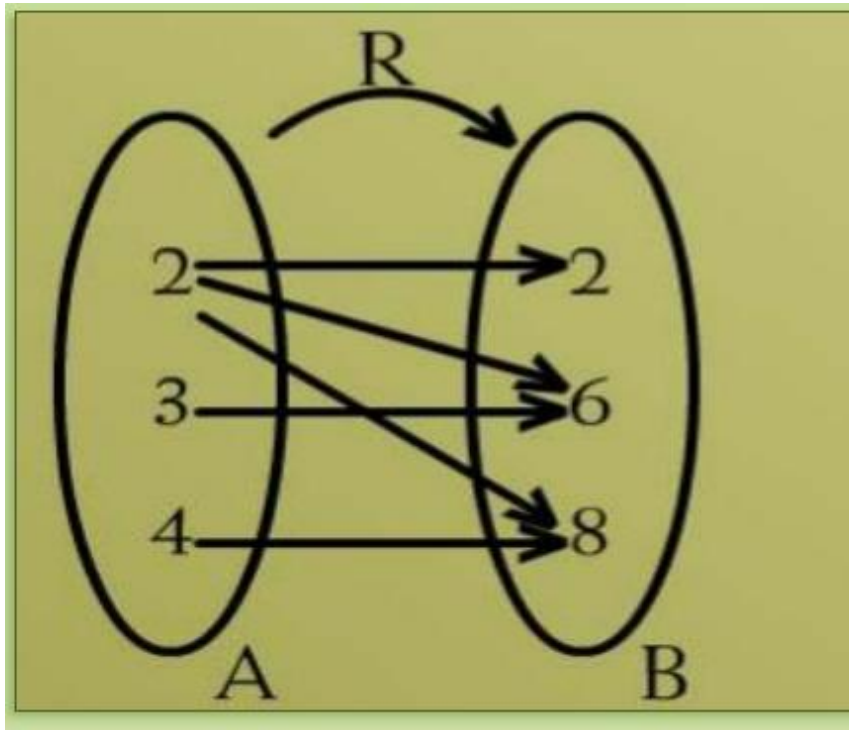
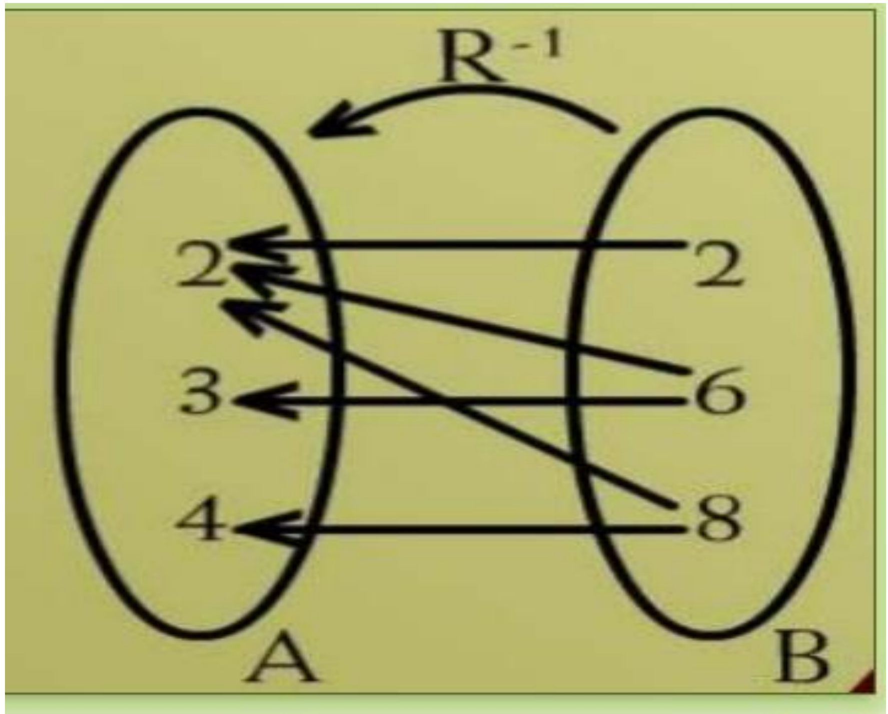
Find the relation a divides b.

R = {(2.2)(2,6)(2,8) (3,6) (4,8)}

$\boldsymbol{R^{-1}}$**= {(2,2) (6,2) (8,2) (6,3) (8,4)}**

Is the inverse of relation $\boldsymbol{R^{-1}}$ ***hold the given condition?***

Arrow Diagram of Inver Relation

R = {(2.2)(2,6)(2,8) (3,6) (4,8)}	R^{-1}= {(2,2) (6,2) (8,2) (6,3) (8,4)}
	

Matrix Representation

R = {*(2.2)(2,6)(2,8) (3,6) (4,8)}*	***R ?***
$M = \begin{array}{cc} & \begin{array}{ccc} 2 & 6 & 8 \end{array} \\ \begin{array}{c} 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$	$M^t = \begin{array}{cc} & \begin{array}{ccc} 2 & 3 & 4 \end{array} \\ \begin{array}{c} 2 \\ 6 \\ 8 \end{array} & \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{array}$

Complementary Relation

- R be a relation from a set A to a set B.

The complementary relation R of R is the set of all those ordered pairs in ***A x B that do not belong to R***.

$$R^c = A \times B - R$$

$$= \{ (a, b) \in A \times B \mid (a, b) \notin R\}$$

Example

- Let A = {1, 2, 3}

  A x A = { (1, 1) , (1, 2) , (1, 3) , (2, 1) , (2, 2) , (2, 3) , (3, 1) , (3, 2) , (3, 3) }

R = { (1, 1) , (1, 3) , (2, 2) , (2, 3) , (3, 1) }

- Then what is R^c.

Example

A x A = { (1, 1) , (1, 2) , (1, 3) , (2, 1) , (2, 2) , (2, 3) , (3, 1) , (3, 2) , (3, 3) }

R = { (1, 1) , (1, 3) , (2, 2) , (2, 3) , (3, 1) }

R^c = { (1, 2), (2, 1) , (3, 2) , (3, 3) }

Composite Relation

- Let A, B and C be three sets.

Suppose that

R is a relation from A to B

S is a relation from B to C.The composite of R and S denoted by S . R is relation from A to CThat consisting the order pair 0f (a,c) where a ∈ A and c ∈ CThere exist an element b ∈ B such (a , b) ∈ R and (b ,c) ∈ .S

$$\boldsymbol{S \cdot R = \{ (a, c) a \in A \text{ and } c \in C , \exists\, b \in B, (a, b) \in R \text{ and } (b,c) \in S\}}$$

Example

Let A= {a,b,c,d} B ={1,2,3,4} and C={x,y,z}

R = {(a,1)(a,4)(b,3)(c,1)(c,4)}

S = {(1,x)(2,x)(3,y)(3,z)}

R . S = (a,x) (b,y)(b,z) (c,x)

Matrix Represntation

Matrix representation of composite relation can be found by taking the Boolean product of the matrices.

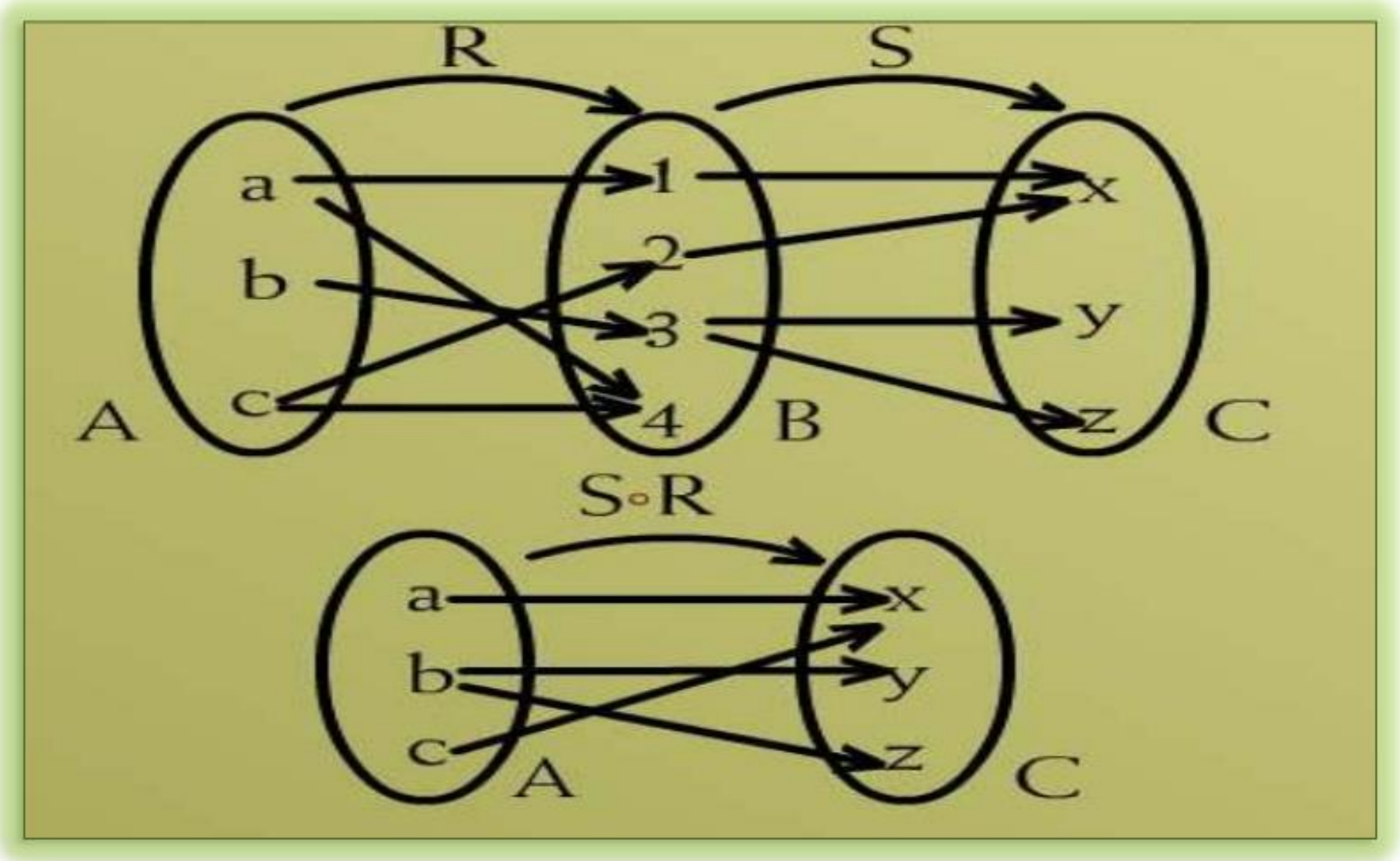
Let we have two matrices Ms and Mr

$$M_s = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad M_r = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Ms . Mr = ?

Arrow Diagram

A = {a, b, c}   C = {x, y, z}



Example

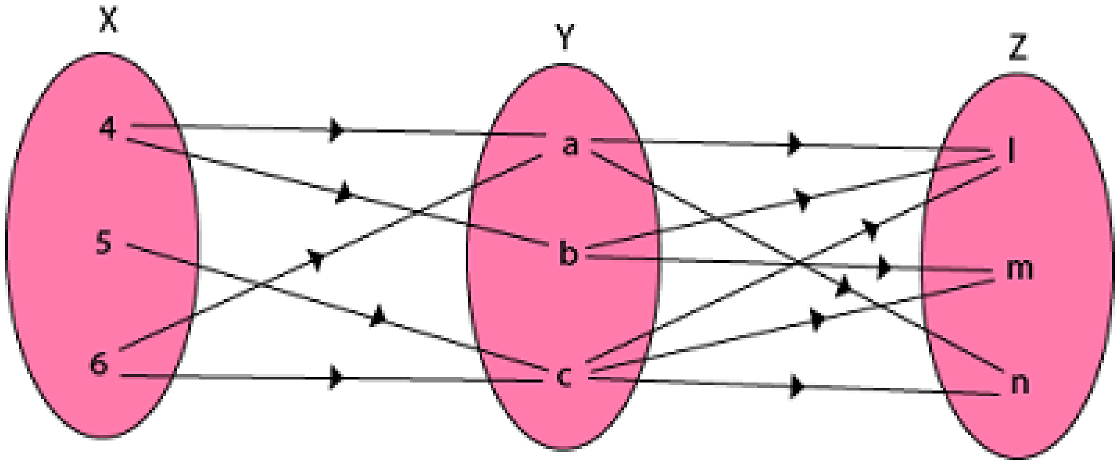
- Let X = {4, 5, 6}, Y = {a, b, c} and Z = {l, m, n}. Consider the relation R R from X to Y and S from Y to Z.

  R = {(4, a), (4, b), (5, c), (6, a), (6, c)}

  S = {(a, l), (a, n), (b, l), (b, m), (c, l), (c, m), (c, n)}

- Find R . S?

Solution



$R \cdot S = = \{(4, l), (4, n), (4, m), (5, l), (5, m), (5, n), (6, l), (6, m), (6, n)\}$